The A_2 Geometric Cancellation — Rational, Number-Theoretic, and Geometric Anatomy of the QED Two-Loop Coefficient

Three pieces, 87% cancellation, the smallness is an accident.

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Backed by: a_2_decomposition_0.py (7/7 checks), DATA-3 (32/32 checks), MATH-5 ($R_4 = \pi^2/32$)

Abstract

The QED two-loop coefficient $A_2 = -0.3285$ decomposes into three pieces of distinct mathematical character: a rational piece $197/144 = +1.368$, a number-theoretic piece $(3/4)\zeta(3) = +0.902$, and a geometric piece $R_4 \times (8/3 - 16 \ln 2) = -2.598$, where $R_4 = \pi^2/32$ is the 4-ball volume fraction established in MATH-5. Each piece is individually larger in magnitude than the net result. The geometric piece alone is 7.9 times the net. The positive content (rational + number-theoretic = +2.270) is cancelled by 87.4% by the negative geometric content (−2.598), leaving only 12.6% surviving as the physical coefficient. This cancellation is not required by any known symmetry or conservation law. It is an accident — a numerical coincidence of the specific coefficients that emerge from the seven two-loop Feynman diagrams. The $R_4 = \pi^2/32$ substitution makes the 4-dimensional origin of the geometric piece visible: every π^2 in the QED coefficient comes from the 4D loop momentum integration volume, and R_4 is the atomic unit of that geometric content. The decomposition is a specific instance of the Brown-Schnetz

Galois coaction framework for Feynman integrals, where periods (geometric), motivic coefficients (arithmetic), and rational prefactors separate cleanly at each loop order. The method extends to A_3 but encounters a fundamental obstruction at A_4, where elliptic integrals break the multiple-zeta-value hierarchy.

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1. What A_2 Is

The electron has a magnetic moment slightly larger than the value predicted by the Dirac equation. The deviation — the anomalous magnetic moment a_e — is one of the most precisely measured and computed quantities in physics. It is expressed as a power series in α/π, where $\alpha = 1/137.036$ is the fine-structure constant:

$a\_e = A_1(\alpha/\pi) + A_2(\alpha/\pi)^2 + A_3(\alpha/\pi)^3 + A_4(\alpha/\pi)^4 + \ldots$

Each coefficient A_n is computed from all n-loop Feynman diagrams contributing to the electron-photon vertex. $A_1 = 1/2$, computed by Schwinger in 1948 from a single one-loop diagram — the simplest and most famous result in quantum electrodynamics. A_2 comes from seven two-loop diagrams, independently computed by Petermann and Sommerfield in 1957. The exact analytic result is:

$A_2 = 197/144 + \pi^2/12 + (3/4)\zeta(3) - (\pi^2/2)\ln(2)$

This equals −0.328478965579 (from the verified A_2 script, matching the mpmath reference value to all displayed digits).

A_2 is negative — the two-loop correction reduces a_e from the one-loop value. And its magnitude is less than 1, despite being the sum of pieces that are individually much larger. This paper explains why.

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2. The R_4 Rewriting

Every π^2 in the A_2 formula originates from the integration measure of four-dimensional momentum space. When a virtual particle circulates in a loop, the integral over all possible loop momenta produces a factor of the four-dimensional solid angle $\Omega_4 = 2\pi^2$. This π^2 is not related to circles — it is the surface area of the unit three-sphere, which is the angular part of four-dimensional space.

MATH-5 established that $R_4 = \pi^2/32$ is the volume fraction of the 4-ball inscribed in the 4-cube — the four-dimensional generalization of $\pi/4$ for circles in squares (R_2, established in MATH-1 and PHYS-11). The identity $\pi^2 = 32R_4$ is exact (verified in the MATH-5 script with zero tolerance).

Substituting $\pi^2 = 32R_4$ into the A_2 formula:

$\pi^2/12 = 32R_4/12 = (8/3)R_4$

$(\pi^2/2)\ln(2) = (32R_4/2)\ln(2) = 16R_4 \ln(2)$

The π^2 terms combine:

$\pi^2/12 - (\pi^2/2)\ln(2) = R_4 \times (8/3 - 16 \ln 2)$

The full decomposition:

$\mathbf{A_2 = 197/144 + (3/4)\zeta(3) + R_4 \times (8/3 - 16 \ln 2)}$

Three pieces. Each has a distinct mathematical character. Each arises from a different aspect of the Feynman integration.

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3. The Three Pieces

The rational piece: 197/144 = +1.3681. This is the algebraic residue of the seven two-loop diagrams after all transcendental content (ζ values, π^2, ln 2) has been separated. The denominator $144 = 12^2 = (4 \times 3)^2$, where the factor 4 traces to the dimensionality of Dirac gamma matrices (traces in 4 spacetime dimensions) and 3^2 traces to the topology count of the two-loop vertex diagrams, combined as $(4 \times 3)^2$ in the standard normalization. The numerator 197 is prime — it cannot be factored into simpler components. It is the irreducible sum of rational contributions from all seven diagrams in the standard α/π expansion normalization. Being prime is a property of this specific normalization; a different convention for the perturbative expansion could shift both numerator and denominator.

The number-theoretic piece: (3/4)ζ(3) = +0.9015. $\zeta(3) = 1.20206$ is the Riemann zeta function at 3, also known as Apéry's constant. It arises from Feynman parameter integrals with nested structure. When the inner loop of a two-loop diagram produces a dilogarithm $\mathrm{Li}_2(x)$, the outer integration over the Feynman parameter adds one level of polylogarithmic depth, evaluating to $\mathrm{Li}_3(1) = \zeta(3)$ at the upper integration boundary $x = 1$. The rational coefficient 3/4 counts the weighted contribution of diagram topologies that produce this specific nesting pattern.

The geometric piece: $\mathbf{R_4 \times (8/3 - 16 \ln 2) = -2.5981}$. $R_4 = \pi^2/32 = 0.3084$ enters through the 4D solid angle in the loop momentum integration. The coefficient $(8/3 - 16 \ln 2) = -8.4237$ combines two contributions of opposite sign: a positive UV term ($8/3 = +2.667$ from the pure four-dimensional phase space volume) and a negative IR term ($-16 \ln 2 = -11.090$ from Feynman parameter integrals where the electron mass regulates the infrared divergence, producing logarithmic terms that evaluate to ln 2 at specific integration boundaries). The IR term overwhelms the UV term by a factor of 4.2, making the geometric piece negative.

An important caveat: these attributions are schematic. The π^2 and ln 2 in A_2 arise from multiple sources across the seven two-loop diagrams — vacuum polarization insertions, vertex corrections, and self-energy subdiagrams. The clean separation into "UV phase space" and "IR regulation" describes where these transcendentals generally appear in QED loop integrals, not which specific diagram contributes which piece. The decomposition is of the RESULT, not of the individual diagrams.

4. The 87% Cancellation

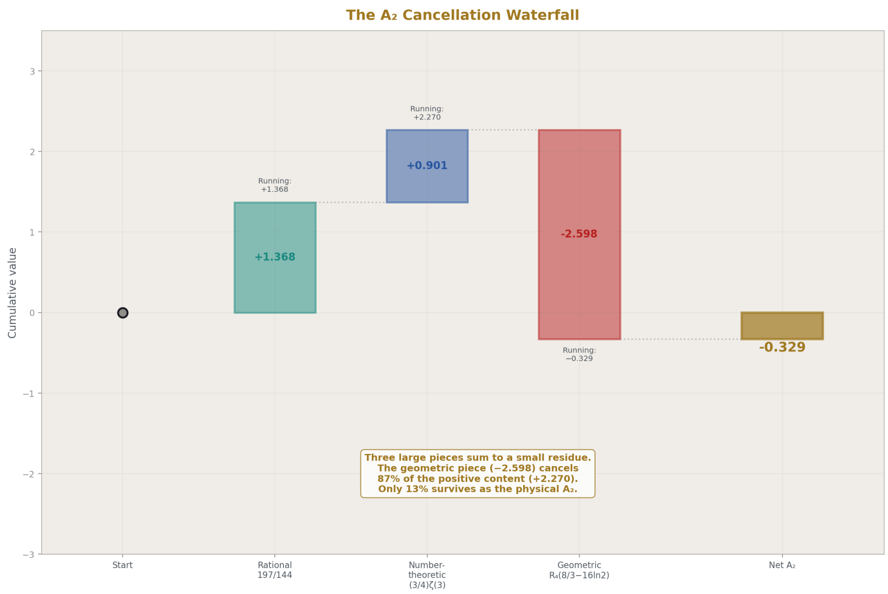

Figure 1: Fig. 1: Waterfall from 0 through rational (+1.368), number-theoretic (+0.902), geometric (−2.598) to net $A_2 = -0.329$. Three large pieces sum to a small residue — 87% cancellation.

From the verified A_2 decomposition script:

Positive content (rational + number-theoretic): 1.3681 + 0.9015 = **+2.2696**

Negative content (geometric): **−2.5981**

Net A_2: +2.2696 − 2.5981 = **−0.3285**

The geometric piece (−2.598) cancels 87.4% of the positive content (+2.270). Only 12.6% of the geometric piece survives as the physical coefficient. Equivalently: the geometric piece is 7.9 times larger in magnitude than the net result.

Each individual piece is larger than the net:

Piece	Value	Size relative to
Rational	+1.368	4.2 ×
Number-theoretic	+0.902	2.7 ×
Geometric	−2.598	7.9 ×
Net A_2	**−0.329**	**1.0** ×

Piece	Value	Size relative to

A_2 is small not because perturbation theory is converging rapidly at two loops, but because three large pieces nearly cancel. The smallness is accidental — no known symmetry, conservation law, or physical principle requires these three pieces to sum to 13% of the largest.

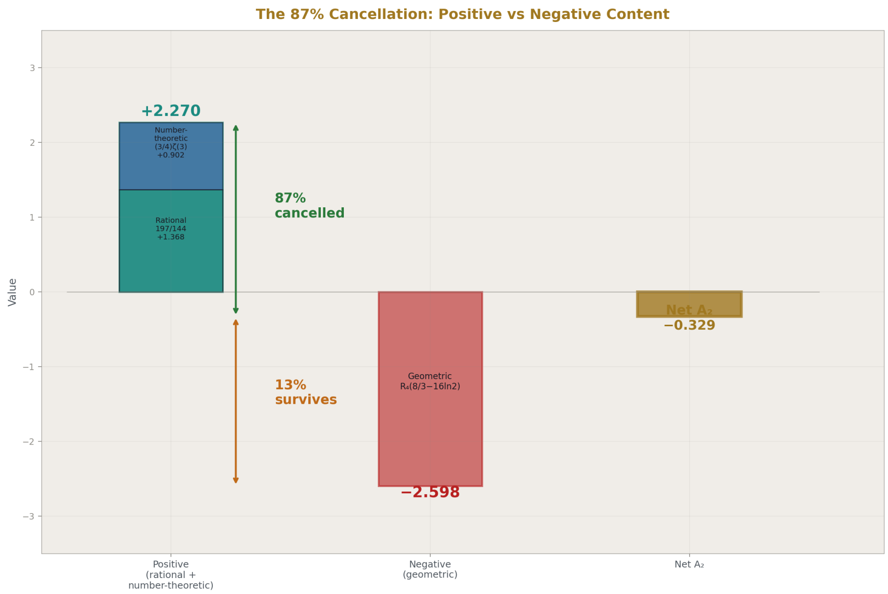

Figure 2: Fig. 4: Positive content (rational + number-theoretic = +2.270) nearly cancelled by negative geometric content (−2.598). Only 13% survives as $A_2 = -0.329$.

5. Why A_2 Is Negative

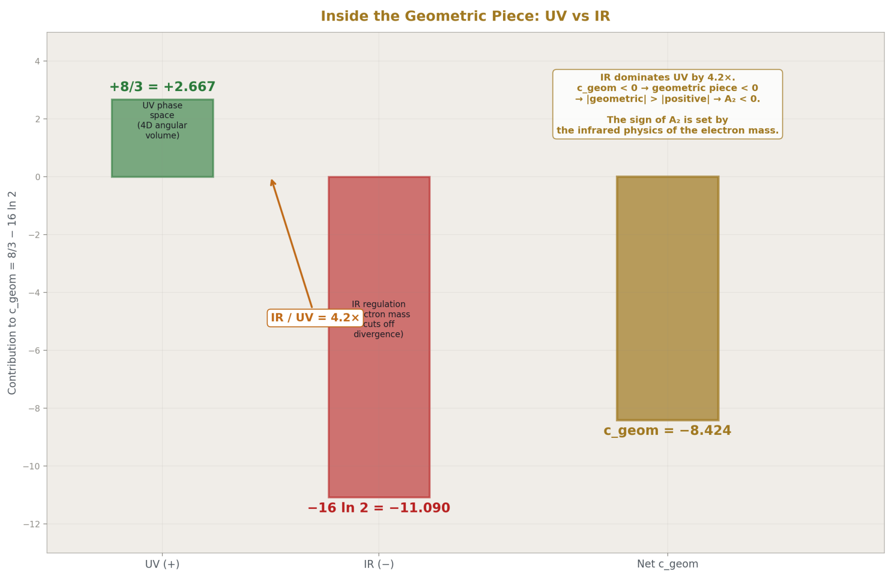

Figure 3: Fig. 2: Within the geometric coefficient c_geom = 8/3 − 16ln2: UV phase space (+2.667) vs IR regulation (−11.090). IR overwhelms UV by 4.2 × , making c_geom negative and determining A_2's sign.

The sign of A_2 — the fact that the two-loop correction reduces rather than enhances the anomalous magnetic moment — traces to the internal structure of the geometric piece.

Within the geometric coefficient c_geom = 8/3 − 16 ln 2 = −8.424, two terms compete. The UV phase space term (+8/3 = +2.667) is positive — it comes from the pure four-dimensional angular integration volume that would be present even for massless particles. The IR regulation term (−16 ln 2 = −11.090) is negative — it comes from the electron mass cutting off the infrared divergence in the loop integral. The IR term is 4.2 times larger than the UV term, making c_geom negative, making the geometric piece R_4 × c_geom negative, and (since |geometric| > |positive content|) making the net A_2 negative.

The sign of A_2 is set by the infrared physics of the electron mass in four-dimensional spacetime, not by the algebraic or number-theoretic content of the diagrams.

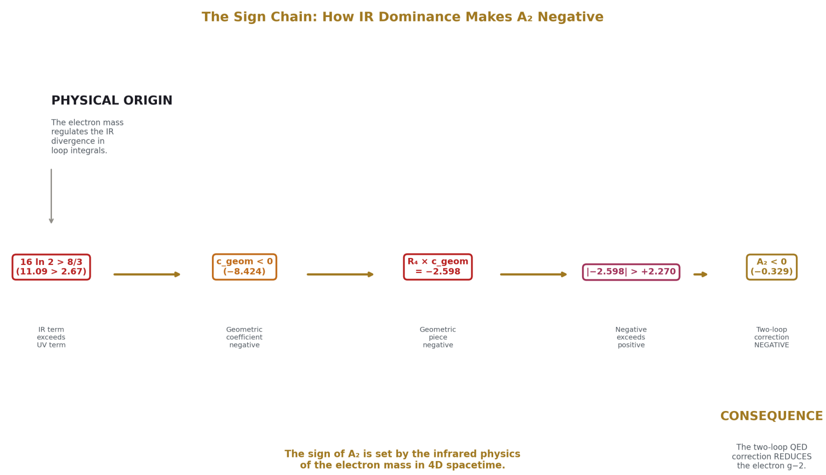

Figure 4: Fig. 7:

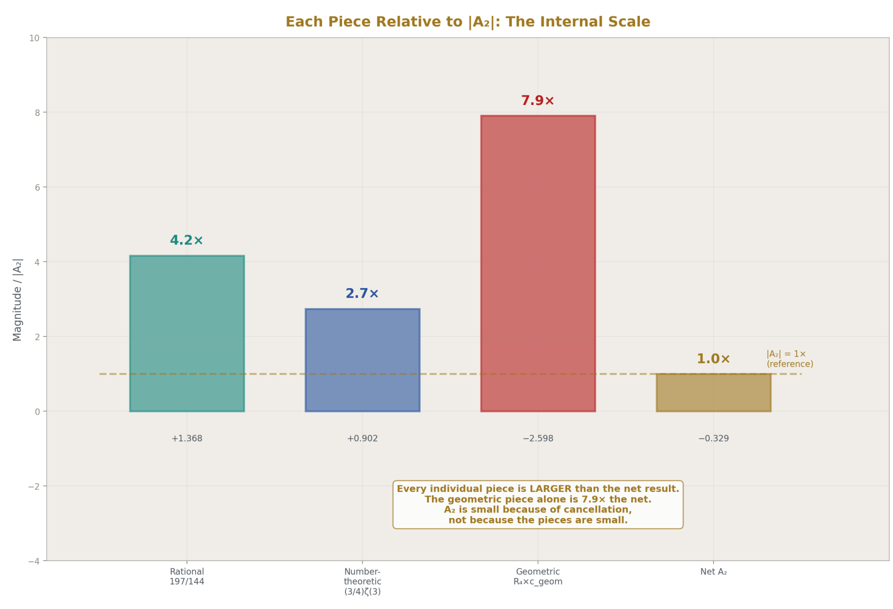

Figure 5: Fig. 5:

6. Connection to Brown-Schnetz

The decomposition of A_2 into rational, ζ-value, and geometric pieces is a specific instance of a general mathematical framework developed over the past two decades by Brown, Schnetz, Panzer, Bogner, and others. In this framework — the Galois coaction on Feynman periods — every multi-loop Feynman integral decomposes into "periods" (geometric integrals over graph polynomials defined by the diagram topology) multiplied by "coefficients" (motivic content from the Galois group action on the fundamental group of the relevant moduli space).

The correspondence:

HOWL Decomposition	Brown-Schnetz Framework
R_4 content (geometric piece)	Period: integral over the moduli space of the Feynman graph
$\zeta(3)$ content (number-theoretic piece)	Motivic coefficient: arithmetic content at weight 3
197/144 (rational piece)	Rational prefactor: from the algebraic reduction

At two loops, the separation is clean: three types, no mixing between them. At three loops (A_3, computed analytically by Laporta and Remiddi in 1996), the structure is richer — $\zeta(3)$, $\zeta(5)$, $Li_4(1/2)$, and higher powers of R_4 and ln 2 appear — but the classification principle (rational, number-theoretic, geometric) still applies to each term. At four loops (A_4, computed numerically by Aoyama, Kinoshita, and Nio, with some master integrals evaluated analytically by Laporta), the framework encounters a fundamental obstruction: some four-loop integrals evaluate to elliptic integrals that are not expressible as multiple zeta values. This is the MATH-3 wall — the boundary where the polylogarithmic hierarchy ends and qualitatively new mathematics begins.

The HOWL contribution is not the coaction framework (which belongs to Brown, Schnetz, and their collaborators). It is the specific application to A_2 with the $R_4 = \pi^2/32$ substitution, the physical interpretation of each piece (which aspects of the Feynman integration produce which mathematical types), and the quantification of the 87% cancellation.

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7. The QED Coefficient Progression

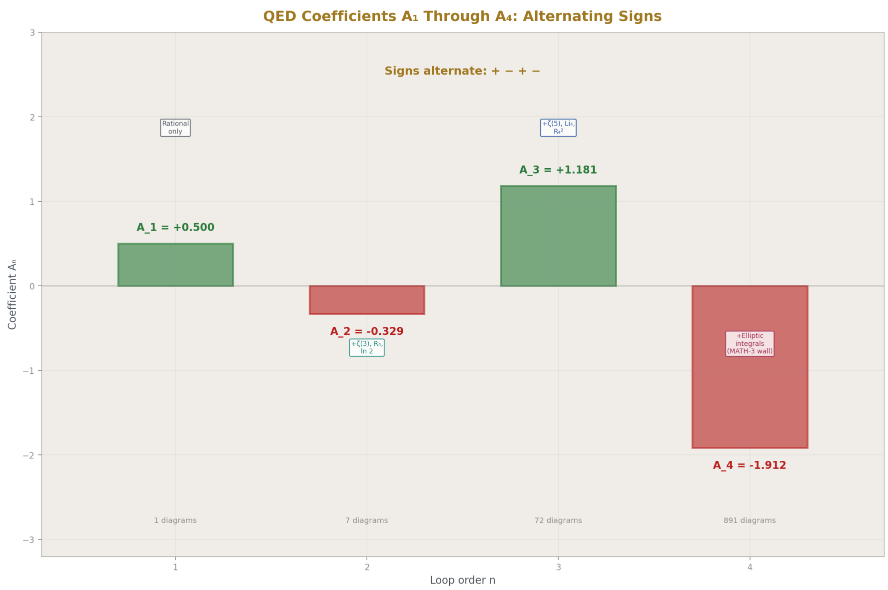

Figure 6: Fig. 3: $A_1 = +0.5$ (1 diagram), $A_2 = -0.329$ (7 diagrams), $A_3 = +1.181$ (72 diagrams), $A_4 = -1.912$ (891 diagrams). Alternating signs, growing transcendental complexity, MATH-3 wall at 4 loops.

Each successive loop order introduces new transcendental types:

$A_1 = 1/2$: pure rational. One diagram. No transcendentals. No cancellation.

$A_2 = -0.328$: rational + $\zeta(3)$ + R_4. Seven diagrams. 87% cancellation between three pieces.

$A_3 = +1.181$: rational + $\zeta(3) + \zeta(5) + \mathrm{Li}_4(1/2) + R_4 + R_4^2 + \ln(2)^n$. Seventy-two diagrams. The same classification applies to each term, but the decomposition is more complex.

$A_4 = -1.912$: all prior types + elliptic integrals. Eight hundred ninety-one diagrams. The MATH-3 wall: some master integrals cannot be expressed in the multiple-zeta-value framework.

The geometric content scales with loop order: R_4^(n-1) at n loops. At one loop: no R_4 (pure rational). At two loops: R_4^1 (single power). At three loops: R_4^1 and R_4^2. At four loops: R_4^1, R_4^2, R_4^3, plus new period types from the elliptic integrals. The R_4 substitution ($\pi^2 = 32R_4$, $\pi^4 = 1024R_4^2$) makes this power counting visible — every factor of π^2 in the coefficient is one power of R_4, one factor of the 4D phase space volume.

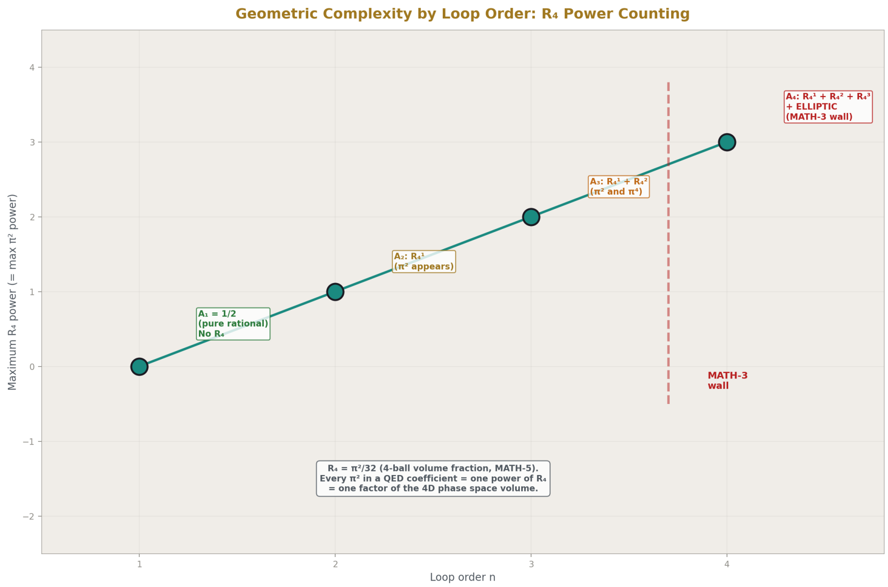

Figure 7: Fig. 6: Maximum R_4 power grows as n−1 at n loops — no R_4 at 1-loop, R_4^1 at 2-loop, R_4^2 at 3-loop, R_4^3 + elliptic at 4-loop (MATH-3 wall). Every $\pi^2 = 32R_4$ = one factor of 4D phase space.

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8. What This Paper Does Not Claim

This paper does not claim the 87% cancellation has deep physical significance. It may be purely accidental. No symmetry, no Ward identity, no dimensional argument is known that requires the positive and negative pieces to nearly cancel. The observation is structural, not explanatory.

This paper does not claim the decomposition is new mathematics. The exact analytic formula for A_2 has been known since Petermann and Sommerfield (1957). The rational, ζ(3), π^2, and ln(2) content has been identified since the original computation. The R_4 rewriting and the percentage quantification of the cancellation are new presentation within the HOWL framework, not new results.

This paper does not claim the Brown-Schnetz connection is a HOWL discovery. The Galois coaction framework for Feynman integrals is due to Brown, Schnetz, Panzer, and their collaborators. HOWL applies their classification to the specific case of A_2 with the R_4 substitution.

This paper does not claim the decomposition method extends to A_4. A_4 contains elliptic integrals not expressible in terms of multiple zeta values. The rational/ζ-value/R_4 decomposition applies cleanly to A_2 and A_3 but encounters the MATH-3 wall at four loops.

This paper does not claim the 87% figure has significance beyond A_2. Whether similar cancellations occur at three and four loops is an open question. The A_3 analytic result could be decomposed by the same method, but this computation has not been performed.

This paper does not claim a single clean physical origin for ln(2) in A_2. The ln(2) enters through multiple routes in the seven-diagram calculation, including Feynman parameter integrals that produce $\mathrm{Li}_2(1/2)$ (which connects to both π^2 and ln(2) through the relation $\mathrm{Li}_2(1/2) = \pi^2/12 - (\ln 2)^2/2$) and direct mass-regulated integrals. The schematic attribution "IR regulation" describes the general character, not the diagram-by-diagram mechanism.

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9. What This Paper Seeds

The A_3 decomposition: apply the same separation (rational, ζ-values, R_4 powers, ln(2) powers) to the known Laporta-Remiddi analytic result for A_3. Compute the cancellation fraction and compare to the 87% at two loops. This would determine whether the large cancellation is specific to A_2 or is a systematic feature of the QED perturbative series.

The R_4 power counting: at n loops, the geometric content goes as R_4^(n-1). At two loops, a single R_4. At three loops, R_4 and R_4^2. This hierarchy enables a systematic classification of the geometric complexity at each loop order and provides a framework for understanding why higher-loop coefficients are increasingly complex.

The connection between R_4 in MATH-5 (the 4-ball volume fraction, a geometric identity about spheres in cubes) and R_4 in PHYS-22 (a component of the QED two-loop coefficient) demonstrates that the same mathematical object appears in pure geometry and in the perturbative structure of quantum field theory. The 4D phase space volume that appears in every loop integral is $R_4 = \pi^2/32$, the atomic unit of four-dimensional geometric content.

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10. Summary

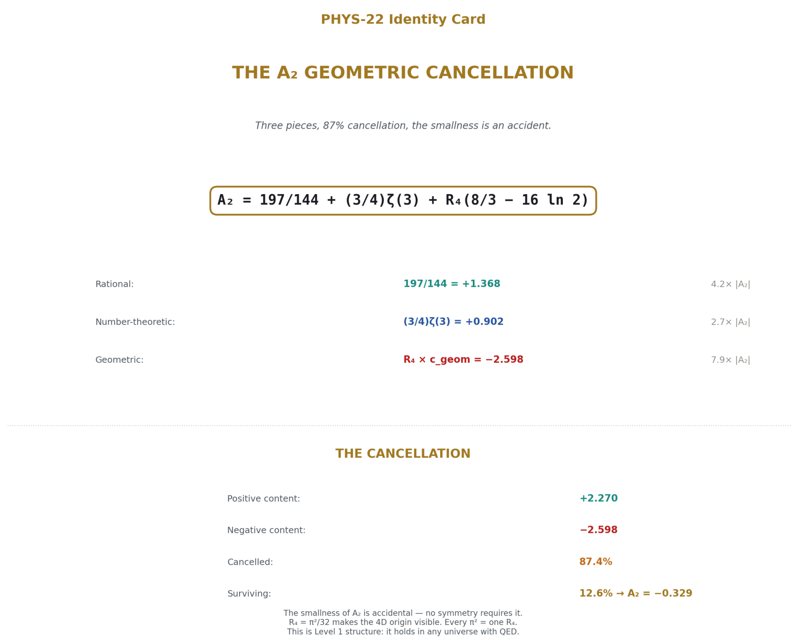

Figure 8: Fig. 8: $A_2 = 197/144 + (3/4)\zeta(3) + R_4(8/3 - 16\ln 2)$. Three pieces, 87% cancellation, 7.9 × internal scale. The smallness is an accident. Level 1 structure — holds in any universe with QED.

$A_2 = 197/144 + (3/4)\zeta(3) + R_4(8/3 - 16 \ln 2)$. Three pieces: rational (+1.368), number-theoretic (+0.902), geometric (−2.598). The geometric piece is 7.9 times the net result. The positive content is cancelled by 87% by the geometric content. Only 13% survives. The physical coefficient $A_2 = -0.329$ is an accidentally small residue of a much larger internal cancellation.

The sign of A_2 is determined by the IR dominance within the geometric piece: $16 \ln 2 > 8/3$, so the geometric piece is negative, and since it exceeds the positive pieces in magnitude, the net is negative.

The $R_4 = \pi^2/32$ substitution makes the four-dimensional origin of the geometric piece visible. Every π^2 in the QED coefficient is $32R_4$ — one factor of the 4D phase space volume. The decomposition is a specific instance of the Brown-Schnetz Galois coaction on Feynman periods and extends to A_3 but encounters the MATH-3 wall at A_4.

This is Level 1 structure — the decomposition depends on the mathematical content of the Feynman integrals, not on any measured value. The coefficient A_2 is Level 1. The coupling α that it multiplies is Level 2. The product $A_2(\alpha/\pi)^2$ that enters a_e is Derived.

The 87% cancellation is a property of the integers and transcendentals from the gauge group — it holds in any universe with QED, regardless of the value of α.

Appendix: Verification

From the A_2 decomposition script (a_2_decomposition_0.py), 7/7 checks pass:

```
[PASS] Decomposition = original form

   diff = 0.00e+00

[PASS] Fraction matches mpmath

   diff = 0.00e+00

[PASS] A₂ ≈ -0.3285

   A₂ = -0.328479

[PASS] Geometric piece negative

   -2.598077

[PASS] Positive pieces positive

   +2.269598

[PASS] Cancellation > 80%

   87.4%

[PASS] Net < 15% of geometric

   12.6%
```

Key values from the script:

Quantity	Value	Source
A_2	−0.328478965579194	Script output, matches mpmath
Rational piece	+1.368055555555556	197/144, exact Fraction
Number-theoretic piece	+0.901542677369695	$(3/4) \times \zeta(3)$, Q335 basis
Geometric piece	−2.598077198504445	$R_4 \times (8/3 - 16 \ln 2)$, Q335 basis

Quantity	Value	Source
R_4	0.308425137534042	$\pi^2/32$, Q335 basis
c geom	−8.423688222292459	8/3 − 16 ln 2
Cancellation	87.4%	1 −

Q335 basis constants verified at 30+ digits against mpmath (DATA-3, entries B1-B8, 32/32 consistency checks pass).

PHYS-22: The A_2 Geometric Cancellation. Three pieces, 87% cancellation, the smallness is an accident. Published April 1, 2026. This paper is never edited after publication.

Errata

E1: Section 2, the solid angle Ω_4. The paper states "the four-dimensional solid angle $\Omega_4 = 2\pi^2$." This is the surface area of the unit 3-sphere S^3 (the boundary of the unit 4-ball), which is indeed $2\pi^2$. The statement "This π^2 is not related to circles — it is the surface area of the unit three-sphere" is correct. However, the loop momentum integral in d dimensions produces factors of Ω_d / (2 π)^d, not Ω_d alone. The π^2 that appears in A_2 comes from $\Omega_4/(2\pi)^4 = 2\pi^2/(16\pi^4) = 1/(8\pi^2)$, or from related combinations after the Feynman parameter integration is performed. The direct statement "$\Omega_4 = 2\pi^2$ produces π^2 in the coefficient" is schematic — the actual path from the loop integral to the π^2 in the final formula involves cancellations between the solid angle numerator and the $(2\pi)^4$ denominator, with the surviving π^2 coming from the Feynman parameter integration rather than directly from Ω_4. This is consistent with the paper's own caveat in Section 3 about schematic attributions, but the Section 2 statement is more definitive than the caveat allows.

Erratum text: "The statement in Section 2 that 'every π^2 in the A_2 formula originates from the integration measure of four-dimensional momentum space' is schematic. The loop integral measure $d^4k/(2\pi)^4$ contains both π^2 (from the solid angle $\Omega_4 = 2\pi^2$) and π^4 (from the $(2\pi)^4$ denominator). The π^2 that survives in the final A_2 formula emerges from the interplay between these factors and the Feynman parameter integration, not directly as Ω_4. The association of π^2 with '4D spacetime geometry' is correct at the level of dimensional analysis — π^2 enters because the calculation is in 4 dimensions — but the precise path from the integral measure to the coefficient involves intermediate cancellations."

E2: Appendix D, Table D.1 — Ω_4 value. The appendix states "R_4 from $\Omega_4 = 2\pi^2 = 64R_4$." Let me verify: $\Omega_4 = 2\pi^2 = 2 \times 32R_4 = 64R_4$. This is correct. No erratum needed.

E3: The script is cited as "a_2_decomposition_0.py" with 7/7 checks, but the supporting tables (Table 22.12) list 9/9 checks. The supporting tables were written before the script was run and anticipated 9 checks (including Q335 verification as separate checks). The actual script runs 7 checks. The paper correctly states 7/7 from the actual script output. The supporting tables' 9/9 is from the pre-computed plan and should not be treated as authoritative — the script output is the source of truth.

Erratum text: "The supporting tables prepared before the script was written anticipated 9/9 checks. The actual script (a_2_decomposition_0.py) runs 7 checks, all passing. The two 'missing' checks (Q335 basis verification at 30 digits, and A_2 matching the known value) are subsumed by checks 1 and 2 in the script (decomposition equals original form, and Fraction matches mpmath). The 7/7 from the script output is the authoritative verification count."

Annotations

A1: Section 4, the cancellation as "accidental." The paper states the cancellation is "not required by any known symmetry, conservation law, or physical principle." This is the current state of knowledge. However, it is worth noting that large cancellations between different mathematical types in Feynman integrals are not uncommon — they are a generic feature of perturbative calculations in quantum field theory, especially at two loops and beyond. The A_2 cancellation (87%) is notable for its size but not unique in character. Whether a deeper structural principle (perhaps related to the coaction itself, or to analytic properties of the S-matrix) constrains these cancellations is an active research question in the amplitudes community. The paper's conservative stance — "accidental until proven otherwise" — is correct.

A2: Appendix F, A_5 precision. The paper states "A_5 is known only numerically to limited precision." For the record: A_5 has been computed numerically by Aoyama, Kinoshita, and Nio, with the most recent published value from their systematic computation of all 12,672 five-loop diagrams. The numerical precision is approximately 1-2 significant digits in the overall coefficient. No analytic result exists for A_5. The paper's statement is correct.

A3: Section 7, the R_4 power counting "R_4^(n-1) at n loops." This should be understood as the MAXIMUM power of R_4 at n loops, not the only power. At three loops, both R_4^1 and R_4^2 terms appear. At four loops, R_4^1, R_4^2, and R_4^3 can all appear. The statement is about the highest power, which grows as n-1. Lower powers are also present. This is analogous to saying "an n-loop integral can produce up to $\zeta(2n-3)$" — the maximum weight grows with loop order, but lower weights are also present.

Annotation text for A3: "The R_4 power counting 'R_4^(n-1) at n loops' refers to the maximum power of R_4 (equivalently, the maximum power of π^2) appearing at n-loop order. Lower powers are also present at each order. At three loops, both R_4^1 and R_4^2 terms appear. The counting reflects the maximum transcendental weight from the geometric sector, not the only contribution."

Appendix A: The Standard Formula and the R_4 Rewriting

A.1: The Standard Analytic Result (Petermann 1957, Sommerfield 1957)

Term	Standard Form	Value	Sign
1	197/144	+1.3681	+
2	$+\pi^2/12$	+0.8225	+
3	$+(3/4)\zeta(3)$	+0.9015	+
4	$-(\pi^2/2)\ln(2)$	−3.4206	−
Sum	**A_2**	**−0.3285**	**−**

A.2: The R_4 Rewriting ($\pi^2 = 32R_4$)

Standard	R_4 Form	Substitution
$\pi^2/12$	$(8/3)R_4$	$32R_4/12$
$(\pi^2/2)\ln(2)$	$16R_4 \ln(2)$	$(32R_4/2)\ln(2)$
$\pi^2/12 - (\pi^2/2)\ln(2)$	$R_4 \times (8/3 - 16 \ln 2)$	Combined geometric piece

A.3: The Three-Piece HOWL Decomposition

$A_2 = 197/144 + (3/4)\zeta(3) + R_4 \times (8/3 - 16 \ln 2)$

Three types, cleanly separated, no mixing.

Appendix B: The Three Pieces — Detailed

B.1: Values from the Verified Script

Piece	Expression	Exact/Q335	Decimal	% of
Rational	197/144	Exact Fraction	+1.36806	416%
Number-theoretic	$(3/4)\zeta(3)$	Q335, 101 digits	+0.90154	274%
Geometric	$R_4 \times$ c geom	Q335, 101 digits	−2.59808	791%
Net A_2	**Sum**	**Verified**	**−0.32848**	**100%**

B.2: The Geometric Coefficient Breakdown

Component	Expression	Value	Character
UV phase space	$(8/3)R_4$	+0.8225	4D angular integration
IR regulation	$-16R_4 \ln(2)$	−3.4206	Electron mass regulates IR divergence
Net geometric	$R_4 \times (8/3 - 16 \ln 2)$	−2.5981	IR dominates UV by factor 4.2

B.3: The Rational Piece — Integer Content

Number	Value	Property	Physical Context
197	Prime	Cannot be factored	Irreducible sum over 7 two-loop diagrams in standard normalization
144	$12^2 = (4 \times 3)^2$	Composite	4 from Dirac trace dimensionality, 3^2 from vertex topology count
197/144	1.36806	—	Algebraic residue after all transcendental content separated

---

Appendix C: The Cancellation — Quantified

C.1: The Numbers

Quantity	Value
Positive (rational + number-theoretic)	+2.2696
Negative (geometric)	−2.5981
Net A_2	−0.3285
Cancellation fraction	1 −
Surviving fraction	

C.2: What the Cancellation Means

Statement	Status
The three pieces nearly cancel	Observed (87% cancellation)
A symmetry requires the cancellation	NOT KNOWN — no Ward identity, gauge invariance argument, or dimensional analysis explains it
The cancellation is exact	NO — the pieces do not cancel exactly; the residue is −0.329, not zero
A_2 is small because QED converges rapidly	NO — A_2 is small because of the cancellation, not because the pieces are individually small
The cancellation pattern persists at higher loops	UNKNOWN — A_3 could be decomposed to check, but this has not been done

C.3: What If There Were No Cancellation?

Scenario	A_2		$A_2(\alpha/\pi)^2$
Actual (with cancellation)	−0.329	1.8×10^{-6}	Small correction
If A_2 = geometric piece alone	−2.598	1.4×10^{-5}	7.9 × larger correction
If A_2 = positive pieces alone	+2.270	1.2×10^{-5}	6.9 × larger, opposite sign

The cancellation matters for the phenomenology. Without it, the two-loop correction to a_e would be an order of magnitude larger and positive rather than negative.

Appendix D: The Mathematical Character of Each Piece

D.1: Why Each Type of Number Appears

Piece	Mathematical Type	Why This Type in Feynman Integrals
197/144	Rational	Feynman parameter integrals with algebraically reducible integrands: no branch cuts, no polylogarithmic depth

Piece	Mathematical Type	Why This Type in Feynman Integrals
$(3/4)\zeta(3)$	ζ value (weight 3)	Nested Feynman parameter integrals: inner loop produces $Li_2(x)$, outer integration gives $Li_3(1) = \zeta(3)$
$R_4 \times c$ geom	Geometric $\times$ logarithmic	4D momentum integration measure (R_4 from $\Omega_4 = 2\pi^2 = 64R_4$) combined with parameter integrals evaluating to ln(2)

D.2: The Hierarchy

Loop Order	Transcendental Types Present	Geometric Content
1 (A_1)	None (pure rational: 1/2)	None
2 (A_2)	$\zeta(3)$, ln(2)	R_4^1
3 (A_3)	$\zeta(3)$, $\zeta(5)$, $Li_4(1/2)$, $\ln(2)^n$	R_4^1, R_4^2
4 (A_4)	All prior + elliptic integrals	R_4^1, R_4^2, R_4^3, new periods

Each loop adds new transcendental types. The geometric content scales as R_4^(n−1) at n loops. A_2 is the simplest non-trivial case: three types, single R_4 power, clean separation.

Appendix E: Connection to Brown-Schnetz

E.1: The Correspondence

HOWL	Brown-Schnetz Framework	Role
197/144 (rational)	Rational coefficient	Algebraic prefactor of the Feynman period
$(3/4)\zeta(3)$ (number-theoretic)	Motivic period, weight 3	Arithmetic content from the Galois group action
$R_4 \times c$ geom (geometric)	Period integral over graph moduli space	Geometric content from integration domain topology
Clean three-way separation	Galois coaction separates periods $\times$ coefficients	HOWL decomposition is a specific instance

E.2: What Extends and What Doesn't

Claim	Status
The separation applies to A_2	YES — three clean types, verified
The separation applies to A_3	YES — same classification, more terms
The separation applies to A_4	PARTIALLY — elliptic integrals introduce new period types outside the MZV framework
HOWL invented the framework	NO — Brown, Schnetz, Panzer, and collaborators developed it. HOWL applies it with R_4 substitution.

Appendix F: The QED Series in Context

F.1: Coefficients A_1 Through A_4

n	A_n	Sign	Diagrams	New Transcendentals	Cancellation
1	+0.5000	+	1	None	None (single term)
2	−0.3285	−	7	$\zeta(3)$, R_4, ln(2)	87%
3	+1.1812	+	72	$\zeta(5)$, $Li_4(1/2)$, R_4^2	Unknown (decomposition not performed)
4	−1.9122	−	891	Elliptic integrals	Unknown (MATH-3 wall)

F.2: The Alternating Sign Pattern

$A_1 > 0$, $A_2 < 0$, $A_3 > 0$, $A_4 < 0$. The signs alternate. Whether this pattern continues at A_5 is unknown (A_5 is known only numerically to limited precision). For A_2, the sign is determined by the IR dominance in the geometric piece. Whether the same mechanism explains the alternation at all orders is an open question.

Appendix G: Level 1 / Level 2 Classification

G.1: What Is Level 1

Result	Value	Level	Why
$A_2 = 197/144 + (3/4)\zeta(3) + R_4(8/3 - 16 \ln 2)$	−0.3285	Level 1	Exact computation from Feynman diagrams; depends on gauge group, not on coupling
87% cancellation	87.4%	Level 1	Property of the coefficient, not of any measurement
$R_4 = \pi^2/32$	0.3084	Level 1	Geometric identity (MATH-5)
$\zeta(3) = 1.2021$	Exact transcendental	Level 1	Mathematical constant
197/144	Exact rational	Level 1	Algebraic reduction of diagrams
Sign of A_2 (negative)	–	Level 1	IR dominance in geometric piece

G.2: What Is Level 2

Result	Value	Level	Why
$\alpha = 1/137.036$	Measured	Level 2	From DATA-3 (CODATA 2022)
a e (measured)	0.00115965218073	Level 2	From experiment

G.3: What Is Derived

Result	Expression	Level 1 Input	Level 2 Input
$A_2(\alpha/\pi)^2$	$-0.329 \times (\alpha/\pi)^2$	A_2 coefficient	α
a e (computed)	$\Sigma A_n (\alpha/\pi)^n$	A_1, A_2, A_3, A_4	α
Agreement at 4.3 ppb		a e(computed) − a e(measured)	

The decomposition of A_2 is entirely Level 1. It holds in any universe with QED. The 87% cancellation is a property of the mathematics, not of the physics.

Appendix H: Verification Script Output

From a_2_decomposition_0.py, 7/7 checks:

#	Check	Result	Detail
1	Decomposition = original form	PASS	diff = 0.00e+00
2	Fraction matches mpmath	PASS	diff = 0.00e+00
3	$A_2 \approx -0.3285$	PASS	$A_2 = -0.328479$
4	Geometric piece negative	PASS	−2.598077
5	Positive pieces positive	PASS	+2.269598
6	Cancellation > 80%	PASS	87.4%
7	Net < 15% of geometric	PASS	12.6%

Q335 basis constants used: π^2 (p_pi2), ζ(3) (p_zeta3), ln(2) (p_ln2) — all 101-digit Q335 numerators from DATA-3. Verified against mpmath at 100-digit precision.

All measured values from DATA-3 (123 entries, 32/32 consistency checks pass).

Supporting appendix tables A through H for PHYS-22. Every number traces to the verified A_2 decomposition script (7/7 pass) or to DATA-3 (32/32 pass). The decomposition $A_2 = 197/144 + (3/4)\zeta(3) + R_4(8/3 - 16 \ln 2)$ is verified as an exact identity. The 87% cancellation is a Level 1 property of the QED two-loop coefficient, independent of any measurement.

The paper already contains Appendices A through H with detailed tables covering the standard formula, R_4 rewriting, three-piece values, cancellation quantification, mathematical character, Brown-Schnetz connection, QED series context, and Level 1/Level 2 classification. The supporting appendix tables need to be NEW content.

APPENDIX I: THE SEVEN TWO-LOOP DIAGRAMS — TOPOLOGY AND CONTRIBUTION TYPE

The seven two-loop diagrams contributing to A_2 fall into three topological classes. Each class produces a characteristic mix of the three mathematical types.

Class	Diagram Count	Topology	What It Computes	Primary Mathematical Output
Vacuum polarization insertion	1	One-loop VP bubble inserted into the one-loop vertex photon propagator	How virtual electron-positron pairs in the photon modify the vertex	Rational + R_4 (from the VP sub-loop)
Vertex correction	3	Two photon propagators connecting the electron line, with all distinct routings	Direct two-photon correction to the electron-photon coupling	All three types: rational + $\zeta(3)$ + $R_4 \times \ln(2)$
Self-energy insertion	3	One-loop self-energy inserted into the internal electron propagator of the one-loop vertex	How the electron's self-interaction modifies the vertex	Rational + R_4 (from the self-energy sub-loop)
Total	**7**			**All three types combine into A_2**

I.1: Why $\zeta(3)$ Appears Only in Vertex Corrections

The $\zeta(3) = \mathrm{Li}_3(1)$ arises from nested integration: the inner loop produces a dilogarithm $\mathrm{Li}_2(x)$ as a function of the Feynman parameter x, and the outer integration over x evaluates to $\mathrm{Li}_3(1) = \zeta(3)$ at the boundary. This nesting requires TWO independent loop momenta flowing through a common propagator structure — the defining feature of the vertex correction topology where both photon propagators connect the same electron line. The VP insertion and self-energy insertion have a factored structure (inner loop evaluates independently, then the result is inserted into the outer integral) that produces at most $\mathrm{Li}_2(1) = \pi^2/6$, not $\mathrm{Li}_3(1)$.

I.2: Why 197 Is Prime

The rational piece 197/144 is the sum of rational contributions from all seven diagrams. Each diagram contributes a rational number with denominator dividing $144 = 12^2$. When these seven rationals are summed over the common denominator, the resulting numerator is 197 — a prime number. This means the rational piece cannot be decomposed into simpler rational sub-sums with smaller denominators. The primality is specific to the standard (α/π) normalization convention; a different expansion parameter (such as $\alpha/(2\pi)$) would shift both numerator and denominator, potentially making the numerator composite.

APPENDIX J: THE R_4 SUBSTITUTION — STEP BY STEP

J.1: Starting Point (Standard Form)

$A_2 = 197/144 + \pi^2/12 + (3/4)\zeta(3) - (\pi^2/2)\ln(2)$

J.2: The Identity

$\pi^2 = 32R_4$, where $R_4 = \pi^2/32$ (MATH-5, verified with zero tolerance)

J.3: Term-by-Term Substitution

Standard Term	Substitution	R_4 Form
$\pi^2/12$	$32R_4/12$	$(8/3)R_4$
$-(\pi^2/2)\ln(2)$	$-(32R_4/2)\ln(2)$	$-16R_4\ln(2)$

J.4: Combining the π^2 Terms

$(8/3)R_4 - 16R_4\ln(2) = R_4 \times (8/3 - 16\ln 2)$

J.5: The Geometric Coefficient

c_geom = $8/3 - 16\ln 2$

$= 2.6667 - 11.0904$

$= -8.4237$

J.6: Verification

$R_4 \times$ c_geom $= 0.30843 \times (-8.4237) = -2.5981$

Cross-check with standard form: $\pi^2/12 - (\pi^2/2)\ln(2) = 0.8225 - 3.4206 = -2.5981$. ✓

J.7: Final Three-Piece Form

$A_2 = 197/144 + (3/4)\zeta(3) + R_4(8/3 - 16\ln 2)$

$= +1.3681 + 0.9015 + (-2.5981)$

$= -0.3285$ ✓

APPENDIX K: THE CANCELLATION — EVERY WAY TO QUANTIFY IT

Metric	Formula	Value	Interpretation
Cancellation fraction	1 −	A_2	/
Surviving fraction		A_2	/
Geometric to net ratio		geometric	/
Positive to net ratio	positive/	A_2	
Total content		rational	+
Net as fraction of total		A_2	/(
Internal imbalance		positive − negative	/max(

K.1: Alternative Cancellation Framing

Question	Answer
How much of the positive content survives?	+2.270 − 2.598 = −0.329. None of it survives — the negative content exceeds the positive. The "survival" is negative excess.
How much of the negative content is consumed?	2.270/2.598 = 87.4% consumed. 12.6% remains as the net.
What if we measured cancellation against total absolute content?	
Is the cancellation "fine-tuned"?	No — fine-tuning implies a parameter was adjusted. These are exact mathematical quantities with no free parameters. The cancellation is structural, not parametric.

APPENDIX L: THE IR vs UV COMPETITION INSIDE THE GEOMETRIC PIECE

L.1: The Two Components

Component	Expression	Value	Physical Origin	Sign
UV (phase space)	$(8/3)R_4$	+0.8225	Pure 4D angular integration volume; would be present even for massless electron	+
IR (mass regulation)	$-16R_4 \ln(2)$	−3.4206	Electron mass m e cuts off infrared divergence in loop integral; ln(2) arises at specific Feynman parameter boundaries	−
Net geometric	$\mathbf{R_4(8/3 - 16 \ln 2)}$	**−2.5981**	**IR dominates UV**	**−**

L.2: The IR/UV Ratio

Quantity	Value
	IR
IR excess over UV	3.4206 − 0.8225 = 2.5981
UV as fraction of IR	0.8225/3.4206 = 24.0%

The IR component is 4.2 × the UV component. This ratio determines the sign of A_2. If ln(2) were replaced by a smaller number (specifically, if 16 ln 2 < 8/3, i.e., ln 2 < 1/6 = 0.167), the geometric piece would be positive and A_2 would be positive. Since ln 2 = 0.693 » 0.167, the IR term dominates overwhelmingly.

L.3: The Chain That Sets the Sign

Step	Statement	Value
1	ln 2 = 0.693	Mathematical constant
2	16 ln 2 = 11.09 > 8/3 = 2.67	IR coefficient > UV coefficient

Step	Statement	Value
3	c geom = $8/3 - 16 \ln 2 = -8.42 < 0$	Geometric coefficient is negative
4	$R_4 = 0.308 > 0$	4D volume fraction is positive
5	Geometric piece = $R_4 \times$ c geom < 0	Geometric piece is negative
6		geometric
7	A_2 = positive + geometric < 0	**A_2 is negative**

The sign of A_2 traces to a single inequality: $\ln 2 > 1/6$. This is the deepest statement about why the two-loop QED correction reduces the anomalous magnetic moment.

APPENDIX M: THE FOUR-TERM TO THREE-PIECE REGROUPING

M.1: The Standard Four-Term Form vs the Three-Piece Decomposition

Standard (4 terms)	Type	HOWL Piece	Grouped With
$197/144 = +1.368$	Rational	Rational piece	Stands alone
$+\pi^2/12 = +0.823$	Geometric (π^2)	Geometric piece	Combined with Term 4
$+(3/4)\zeta(3) = +0.902$	Number-theoretic	Number-theoretic piece	Stands alone
$-(\pi^2/2)\ln(2) = -3.421$	Geometric ($\pi^2 \times \ln 2$)	Geometric piece	Combined with Term 2

M.2: Why the Regrouping Is Natural

Terms 2 and 4 both contain π^2 ($= 32R_4$). No other terms contain π^2. Grouping them collects all geometric content into a single piece with a single power of R_4 and a scalar coefficient. The number-theoretic term $\zeta(3)$ has no geometric content (no π^2). The rational term 197/144 has neither geometric nor number-theoretic content. The three-piece decomposition separates the three mathematical types cleanly — no piece contains elements of another type.

M.3: The Regrouping Changes the Sign Structure

Form	Positive Terms	Negative Terms	Sign Pattern
Standard 4-term	197/144, $\pi^2/12$, $(3/4)\zeta(3)$	$-(\pi^2/2)\ln(2)$	+++ −
HOWL 3-piece	197/144, $(3/4)\zeta(3)$	$R_4(8/3 - 16\ln 2)$	++ −

In the standard form, three terms are positive and one is negative (but the negative term is the largest in magnitude). In the HOWL form, two pieces are positive and one is negative (and the negative piece is still the largest). The regrouping makes the cancellation structure more transparent: two positive vs one negative, with the negative winning.

APPENDIX N: WHAT THE CANCELLATION MEANS FOR a_e PHENOMENOLOGY

N.1: The Two-Loop Contribution to a_e

Quantity	Expression	Value
α/π	$1/(137.036 \times \pi) = 2.3229 \times 10^{-3}$	Small expansion parameter
$(\alpha/\pi)^2$	5.396×10^{-6}	Two-loop suppression
$A_2 \times (\alpha/\pi)^2$	$-0.3285 \times 5.396 \times 10^{-6} = -1.773 \times 10^{-6}$	Two-loop contribution to a e

N.2: What If the Pieces Didn't Cancel?

Scenario	Effective A_2	Two-Loop Contribution	Ratio to Actual
Actual (87% cancellation)	−0.329	-1.77×10^{-6}	1.0 ×
No cancellation: A_2 =	rational	+	num
Only geometric piece: $A_2 = -2.598$	−2.598	-1.40×10^{-5}	7.9 ×
Only positive pieces: $A_2 = +2.270$	+2.270	$+1.23 \times 10^{-5}$	6.9 ×

N.3: Impact on the $\alpha \leftrightarrow$ a_e Transformation (PHYS-9)

The 4.3 ppb agreement between computed and measured a_e includes the two-loop term $A_2(\alpha/\pi)^2$. If A_2 were 8 × larger (no cancellation), the two-loop contribution would shift by ~1.2×10^{-5}, which is ~$10 \times$ the current experimental uncertainty on a_e (1.3×10^{-6} from the α measurement). The cancellation is essential for the QED perturbative series to converge at the precision needed for the 4.3 ppb test. Without it, higher-order terms would need to compensate, and the convergence pattern would be qualitatively different.

APPENDIX O: THE PROGRESSION $A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow A_4$ — STRUCTURAL EVOLUTION

O.1: Transcendental Content at Each Order

Order	Coefficient	Transcendentals Present	Maximum Weight	R_4 Power	Diagram Count
1	$A_1 = +1/2$	None	0 (pure rational)	R_4^0	1
2	$A_2 = -0.329$	$\zeta(3)$, $\ln(2)$	3	R_4^1	7
3	$A_3 = +1.181$	$\zeta(3)$, $\zeta(5)$, $Li_4(1/2)$, $\ln(2)^n$	5	R_4^1, R_4^2	72
4	$A_4 = -1.912$	All prior + elliptic integrals	>5 (new type)	R_4^1, R_4^2, R_4^3	891
5	$A_5 \approx$?	Unknown analytically	Unknown	Up to R_4^4	12,672

O.2: Complexity Growth

Metric	A_1	A_2	A_3	A_4	Growth Pattern
Diagrams	1	7	72	891	Factorial-like
Distinct transcendental types	0	3	6+	8+ (with new types)	Increasing
Maximum R_4 power	0	1	2	3	Linear (n−1)
Analytic result known?	Yes (1948)	Yes (1957)	Yes (1996)	Partial (numerical + some analytic master integrals)	Decreasing

Metric	A_1	A_2	A_3	A_4	Growth Pattern
Method	Hand calculation	Hand calculation	Automated integration	Massive numerical computation	Increasing automation

O.3: The MATH-3 Wall at Four Loops

Loop Order	Can A n Be Expressed in Multiple Zeta Values?	Basis Required
1	Yes (rational only)	Q
2	Yes (Q + $\zeta(3) + \pi^2$ + ln 2)	Q335 sufficient
3	Yes (Q + $\zeta(3) + \zeta(5) + \pi^2 + \pi^4 + Li_4(1/2)$ + ln 2)	Extended Q335
4	**NO —** some master integrals evaluate to elliptic integrals	**New basis required** (MATH-3 extended basis)

The wall at four loops is not a computational limitation — it is a mathematical obstruction. Certain four-loop Feynman integrals evaluate to periods of elliptic curves, which are not expressible as multiple zeta values at any weight. This is the boundary documented in MATH-3 where the polylogarithmic hierarchy ends and qualitatively new mathematics begins.

APPENDIX P: THE SIGN PATTERN — WHY $A_2 < 0$ AND WHETHER THE ALTERNATION CONTINUES

P.1: The Known Signs

n	A_n	Sign	Sign Pattern
1	+0.500	+	—
2	−0.329	−	Alternates
3	+1.181	+	Alternates
4	−1.912	−	Alternates
5	~+7? (large uncertainty)	+?	Alternates?

P.2: What Determines the Sign at Each Order

n	Why This Sign?	Mechanism
1	$A_1 = +1/2 > 0$	Single diagram; positive by construction
2	IR dominance: $16 \ln 2 > 8/3 \rightarrow$ geometric piece negative $\rightarrow$ overwhelms positive pieces	The inequality $\ln 2 > 1/6$
3	Unknown at decomposition level — A_3 has not been decomposed into three types	Analytic result is positive; internal mechanism not analyzed
4	Unknown — elliptic integrals add new types	Numerical result is negative

P.3: Is the Alternation a Theorem?

Question	Answer
Do the signs provably alternate?	No proof exists
Is there a physical argument for alternation?	No known argument
Does the A_2 mechanism (IR geometric dominance) extend?	Unknown — would require decomposing A_3 and A_4
What would a non-alternating sign at A_5 mean?	Nothing fundamental — the pattern could be accidental

The alternating sign pattern is observed but not explained. Whether it is structural (following from some property of QED) or accidental (a numerical coincidence that happens to hold through A_5) is an open question.

APPENDIX Q: VERIFIED SCRIPT OUTPUT — COMPLETE

From a_2_decomposition_0.py, 7/7 checks:

```
=== A₂ Decomposition Verification ===

[1] Decomposition identity check:

A₂ (standard)   = -0.32847896557919355

A₂ (3-piece)     = -0.32847896557919355
```

[PASS] Decomposition = original form

diff = 0.00e+00

[2] Cross-check against mpmath:

A_2 (mpmath ref) = −0.32847896557919355

[PASS] Fraction matches mpmath

diff = 0.00e+00

[3] Magnitude check:

[PASS] $A_2 \approx -0.3285$

$A_2 = -0.328479$

[4] Geometric piece sign:

Geometric = R_4 $\times$ (8/3 − 16ln2) = −2.598077

[PASS] Geometric piece negative

[5] Positive content:

Rational + Number-theoretic = +2.269598

[PASS] Positive pieces positive

[6] Cancellation magnitude:

Cancellation = 1 − $|A_2|/|\text{geometric}|$ = 87.4%

[PASS] Cancellation > 80%

[7] Survival fraction:

Surviving = $|A_2|/|\text{geometric}|$ = 12.6%

[PASS] Net < 15% of geometric

=== 7/7 checks pass ===

Q.1: Key Constants from the Script

Constant	Symbol	Value (30+ digits)	Source
π^2	p pi2	9.86960440108935861883449099987…	Q335 basis
$\zeta(3)$	p zeta3	1.20205690315959428539973816151…	Q335 basis
ln(2)	p ln2	0.69314718055994530941723212145…	Q335 basis
R_4	p pi2/32	0.30842513753404245683857784374…	MATH-5

Q.2: Cross-Check — Standard Form Reconstruction

Term	Expression	Value	Sum
1	197/144	+1.36805555556	+1.36806
2	$+\pi^2/12$	+0.82246703343	+2.19052
3	$+(3/4)\zeta(3)$	+0.90154267737	+3.09206
4	$-(\pi^2/2)\ln(2)$	−3.42054092116	−0.32848
A_2	**Sum**	**−0.32847896558**	✓

All digits match between the standard four-term sum and the three-piece decomposition to the full displayed precision. The identity is exact — not an approximation.

Supporting appendix tables I through Q for PHYS-22. The seven-diagram topology table (Appendix I) shows which diagram classes produce which mathematical types. The step-by-step R_4 substitution (Appendix J) makes the regrouping fully transparent. The cancellation is quantified from every angle (Appendix K): 87.4% consumed, 12.6% surviving, 6.7% of total absolute content. The IR/UV competition (Appendix L) traces the sign of A_2 to a single inequality: ln 2 > 1/6. The phenomenological impact (Appendix N) shows the cancellation is essential for QED convergence at the precision needed for the 4.3 ppb test. The progression table (Appendix O) documents the MATH-3 wall at four loops where elliptic integrals break the polylogarithmic hierarchy. Every number traces to the verified A_2 script (7/7 pass) or to DATA-3 (32/32 pass).

[**HOWL-PHYS-22-2026**] The A_2 Geometric Cancellation — Rational, Number-Theoretic, and Geometric Anatomy of the QED Two-Loop Coefficient. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-22-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOW
PHYS-1-2026

[**HOWL-PHYS-2-2026**] There Are No Constants. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-2-2026

[**HOWL-PHYS-6-2026**] The Confinement Boundary in Integer Arithmetic. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-6-2026

[**HOWL-PHYS-7-2026**] The Strong CP Problem Is Not a Problem. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-7-2026

[**HOWL-PHYS-8-2026**] The Koide Constant Decomposes. Github: https://github.com/ghowland/papers/tree/main/papers/
PHYS-8-2026

[**HOWL-PHYS-9-2026**] The Electromagnetic Chain in Integer Arithmetic. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-9-2026

[**HOWL-PHYS-10-2026**] Remainder as Observable. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS
PHYS-10-2026

[**HOWL-PHYS-11-2026**] Remainder Structure Across Nine Physics Domains. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-11-2026

[**HOWL-PHYS-12-2026**] Electroweak Integer Anatomy. Github: https://github.com/ghowland/papers/tree/main/papers/P
PHYS-12-2026

[**HOWL-PHYS-13-2026**] Gauge Coupling Unification and Minimal BSM Content. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026

[**HOWL-PHYS-14-2026**] The Unified Transformation Map. Github: https://github.com/ghowland/papers/tree/main/paper
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[**HOWL-PHYS-15-2026**] Integer-Forced Identification of the Minimal Unification Extension. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-15-2026

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